Power Series Distribution

1 Definition

Now, consider the power series $\sum_x a(x)\theta^x$, let, $\sum a(x)\theta^x$ be convergent for $-p\theta < p, \sum_x a(x)\theta^x = g(\theta)$ for $0 \leqslant \theta < p$.

$$\sum_x \frac{a(x)\theta^x}{g(\theta)} = 1 \quad \text{and also} \quad \frac{a(x)\theta^x}{g(\theta)} \geqslant 0 \quad \forall x = 0, 1, 2, \ldots.$$

Hence, $\left\{ \frac{a(a)\theta^x}{g(\theta)}, x = 0, 1, 2, \ldots \right\}$ describes a PMF and the probability distribution is known as Power series.

1. Set $g(x) = \frac{1}{x!}, x = 0, 1, 2, \ldots$

$\sum_{x=0}^{\infty} \frac{\theta^x}{x!} = e^{\theta} \Rightarrow f(x) = \frac{e^{-\theta}\theta^x}{x!}$. It is a Poisson Distribution.

2. Set $a(x) = \begin{cases} \binom{n}{x} & x = 0, 1, 2, \ldots \\ 0 & \text{otherwise} \end{cases}$,

$$\sum_{x=0}^{\infty} a(x)\theta^x = \sum_{x=0}^{\infty} \binom{n}{x}\theta^x = (1+\theta)^n$$

$\therefore f(x) = \binom{n}{x}\frac{\theta^x}{(1+\theta)^n} = \binom{n}{x}\left(\frac{\theta}{1+\theta}\right)^x \left(1 - \frac{\theta}{1+\theta}\right)^{n-x} <$ It is a Binomial Distribution.

3. Set $a(x) = \binom{x+r-1}{x}, x = 0, 1, 2, \ldots$

$$\sum_x a(x)\theta^x = \sum_x \binom{x+r-1}{x}\theta^x = (1-\theta)^{-r}$$

$f(x) = \binom{x+r-1}{x}(1-\theta)^r\theta^x$ It is a Negative Binomial Distribution.

4. Set $a(x) = 1, \quad \sum_{x=0}^{\infty} a(x)\theta^x = (1-\theta)^{-1}$ for $|\theta| < 1 \Rightarrow -1 < \theta < 1$

$f(x) = (1-\theta)\theta^x, x = 0, 1, 2, \ldots$ It is a Geometric Distribution.

- It is important to note that the power series distribution includes Binomial, Poisson, Negative Binomial, Geometric distributions etc.

2 Mean and Variance

Here, $g(\theta) = \sum_{x=0}^{\infty} a(x)\theta^x, 0 < \theta < p$.

Differentiating w.r.t $\theta, g'(\theta) = \sum_x a(x)x\theta^{x-1}$

Again differentiating w.r.t $\theta, g''(\theta) = \sum_x a(x)x(x-1)\theta^{x-2}$

$$E(X) = \sum_x x\frac{a(x)\theta^x}{g(\theta)} = \frac{\theta}{g(\theta)}\sum_x xa(x)\theta^{x-1} = \frac{\theta}{g(\theta)}g'(\theta) = \theta\frac{d}{d\theta}[\ln g(\theta)]$$

$$E(X(X-1)) = \sum_x x(x-1)\frac{a(x)\theta^x}{g(\theta)} = \frac{\theta^2}{g(\theta)}\sum_x a(x)x(x-1)\theta^{x-2} = \frac{\theta^2}{g(\theta)}g''(\theta)$$

$$\begin{aligned}
&\text{Var}(x) = E(X(X-1)) + E(X) - E^2(X) \\
&= \frac{\theta^2}{g(\theta)} g''(\theta) - \left(\frac{\theta g'(\theta)}{g(\theta)}\right)^2 + E(X) \\
&= \frac{\theta^2 g''(\theta) g(\theta) - \theta^2 \left(g'(\theta)\right)^2}{(g(\theta))^2} + E(X) \\
&= \theta^2 \frac{d^2}{d\theta^2} [\ln g(\theta)] + [E(X)]
\end{aligned}$$

1. A sum related to concept discussed over here. If $f(x)f'(x) < 0, \forall x \in R$
(1) $f(x)$ is monotonically increasing.
(2) $f(x)$ is monotonically decreasing.
(3) $|f(x)|$ is monotonically increasing.
(4) $|f(x)|$ is monotonically decreasing.

$$\begin{aligned}
&f(x)f'(x) < 0. \\
\Rightarrow &\frac{f(x)f'(x)}{f(x)f(x)} < 0 \Rightarrow \frac{f'(x)}{f(x)} < 0 \Rightarrow \frac{d}{dx}(\ln |f(x)|) < 0.
\end{aligned}$$

$\therefore \quad lnf(x) \mid$ is decreasing.

$$\Rightarrow |f(x)| \text{ is decreasing.}$$

2. In a class of power series distribution mean=variance iff the distribution is Poisson.

If Part

Let X be a random variable having a power series distribution with pmf is given by $f(x) = \begin{cases} \frac{a(x)\theta^x}{g(\theta)} & x = 1, 2, \ldots \\ 0, & \text{ow} \end{cases}$

$$g(\theta) = \sum_x a(x)\theta^x, 0 < \theta < p$$

$$\begin{aligned}
&\text{Now, } E(X) = V(X) = \theta^2 \frac{d^2}{d\theta^2} ln(g(\theta)) + E(X) \\
&\Rightarrow E(X) = \frac{d^2}{d\theta^2} \ln(g(\theta)) = 0 \\
&\Rightarrow \ln(g(\theta)) = k\theta + m \Rightarrow g(\theta) = e^m e^{k\theta} = ce^{k\theta} \\
&\Rightarrow c = e^m \\
&\sum_x a(x)\theta^x = c \sum_{x=0}^{\infty} \frac{(k\theta)^x}{x!} \forall \theta \in (0, p)
\end{aligned}$$

Equating the efficient of $\theta^x, \quad a(x) = c\frac{k^x}{x!}$

$$f(x) = \frac{c\frac{k^x}{x!}\theta^x}{ce^{k\theta}} = e^{-k\theta}\frac{(k\theta)^x}{x!}$$

So, $X \sim P(\lambda = k\theta)$

Only if Part

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\begin{aligned}
&\text{If } X \sim \text{ Poisson } (\theta)\\
&g(\theta) = \sum_x a(x)\theta^x = \sum_x \frac{1}{x!}\theta^x = e^\theta\\
&E(X) = \theta\frac{d}{d\theta}\left(\ln e^\theta\right) = \theta\frac{1}{e^\theta}e^\theta = \theta\\
&V(X) = \theta^2\frac{d^2}{d\theta^2}\left[\ln e^\theta\right] + \theta = \theta^2\frac{d}{d\theta}\left[\frac{1}{e^\theta}e^\theta\right] + \theta = \theta^2\cdot 0 + \theta = \theta\\
&\therefore E(X) = \text{Var}(X)
\end{aligned}
$$

3 Recursion relation for central moment

Kth odder central moment of a R.V X is $E(X - E(X)^k = \mu_k$, If $X \sim$ power Series Distribution, $f(x) = \frac{a(x)\theta^x}{g(\theta)}, x = 0, 1, 2, \ldots$

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\begin{aligned}
a(x) \geqslant & 0, \forall x \text{ and } \sum a(x) = 1\\
E(X) = & \theta\frac{d}{d\theta}[\ln g(\theta)]\\
\mu_k = & \sum_{x=0}^{\infty}\left[x - \theta\frac{d}{d\theta}\ln g(\theta)\right]^k \frac{a(x)\theta^x}{g(\theta)}\\
\frac{d\mu_k}{d\theta} = & \sum_{x=0}^{\infty} k\left(x - \theta\frac{d}{d\theta}\ln g(\theta)\right)^{k-1}(-1)\left\{\theta\frac{d^2}{d\theta^2}\ln g(\theta) + \frac{d}{d\theta}\ln g(\theta)\right\}\cdot\frac{a(x)\theta^x}{g(\theta)}\\
& + \sum_{x=0}^{\infty}\left(x - \theta\frac{d}{d\theta}\ln g(\theta)\right)^k a(x)\left\{\frac{g(\theta)\theta^{x-1}x - \theta^x g'(\theta)}{g^2(\theta)}\right\}
\end{aligned}
$$

$$
\begin{aligned}
\Rightarrow \frac{d\mu_k}{d\theta} = & (-k)\left\{\theta\frac{d^2}{d\theta^2}\ln g(\theta) + \frac{d}{d\theta}\ln g(\theta)\right\}\mu_{k-1} + \sum_{x=0}^{0}\left(x - \theta\frac{d}{d\theta}\ln g(\theta)^k\theta^{x-1}a(x)\left\{\frac{g(\theta)x - \theta g'(\theta)}{g^2(\theta)}\right\}\right.\\
= & (-k)\left\{\theta\frac{d^2}{d\theta^2}\ln g(\theta) + \frac{d}{d\theta}\ln g(\theta)\right\}\mu_{k-1} + \frac{1}{\theta}\Sigma(x - E(X))^k\frac{\theta^x(a(x))}{g(\theta)}\left\{x - \frac{\theta g'(\theta)}{g(\theta)}\right\}\\
\frac{d}{d\theta}\mu_k = & (-k)\left\{\theta\frac{d^2}{d\theta^2}\ln g(\theta)^2 + \frac{d}{d\theta}\ln g(\theta)\right\}\mu_{k-1} + \frac{1}{\theta}\mu_{k+1}
\end{aligned}
$$

4 Quiz

1. Let $f(x) = \frac{2000}{\Gamma x+1\Gamma 2000-x+1}\left(\frac{1}{3}\right)^x\left(\frac{2}{3}\right)^{-x}, x = 0, 1, 2, \ldots 2000$. Find the maximum value for the function.
   We know,

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\begin{aligned}
&\sum f(x)c = 1\\
&\Rightarrow \sum_{x=0}^{2000} cf\frac{2000}{\Gamma x + 1\Gamma 2000 - x + 1}\left(\frac{1}{3}\right)^x\left(\frac{2}{3}\right)^{-x} = 1\\
&\Rightarrow \sum_{x=0}^{2000}\left(\frac{2}{3}\right)^{2000}\frac{2000!}{x!(2000-x)!}\left(\frac{1}{3}\right)^x\left(\frac{2}{3}\right)^{2000-x} = 1\\
&\therefore x \sim \text{Bin}\left(2000, \frac{1}{3}\right).\\
&\text{Maximum value } = \text{ mode } = (n+1)p = (2000+1)\frac{1}{3}
\end{aligned}
$$

2. If $X \backsim \text{Bin}\left(2021, \frac{1}{2}\right)$, Find the value of $P(X \leqslant 1010) = I_{1/2}(2021-1010, 1010+1) = I_{1/2}(1011, 1011)$.
$\Rightarrow P(X < 1011) = \frac{1}{2}$

3. If $X_1 \sim \text{Bin}\left(5, \frac{1}{2}\right), X_2 \sim \text{Bin}\left(7, \frac{1}{2}\right)$. So, distribution of $(X_1 + X_2)$ =?
The condition of independence of X_1 and X_2 are not given, so, reproductive property does not hold.

4. $f(b) = (2020 \times 2019) \times \frac{1}{100} - 404b + b^2 + 202$. Find which value of b for which $f(b)$ maximises.

$$\begin{aligned}
&= E\left(X(X-1) - 2E(X)b + b^2 + E(X)\right) \\
&= E\left(X^2\right) - 2bE(X) + b^2 \\
&= E(X-b)^2 \geqslant E(X-\mu)^2 \quad \therefore \quad 4 = np = 202
\end{aligned}$$

5. If X_1, X_2, X_3, X_4 are distributed an absolutely continuous CDF $F(\cdot)$. What is $P\left(X_1 < X_2 > X_3 < X_4\right)$.
For continuous distribution, $f\left(x_1 = x_2 = x_3 = x_4\right)$ or any $p\left(x_1 = x_2\right) = 0$.

$$\therefore P\left(X_1 < X_2 > X_3 < X_4\right) = \frac{1}{24}$$

6. Expectation of matching problem is always 1 irrespective the value of n. (Remind about the technique of getting expectation as a sum).

7. If $x_i \sim \text{Geo}\left(p_i\right)$ then find $\sum X_i \sim$?
It can't follow, $\text{NB}(r, \phi)$ as p_i's are not constant. We need equal success probability to get NB from Geometric Distribution.

8. If $X_i \sim \text{Geo}(p_i)$, find the distribution of $Y = \min\{X_i\}$.

$$\begin{aligned}
\min\{X_i\} &= Y \\
F_Y(y) = p[Y \leqslant y] &= 1 - p\left[Y > y\right] \\
&= 1 - p\left[\min X \mid > y'\right] \\
&= 1 - p\left[X_1 > y, X_2 > y_1, \ldots, X_n > y\right] \\
&= 1 - \prod_{i=1}^{n} p\left(X_1 > y\right) \\
&= 1 - \prod_{i=1}^{n} q_i^{y+1} \\
&= 1 - \prod_{i=1}^{n} \left(1 - p\left(X_i \leq y\right)\right) \\
P(Y = y) = F(y) - F(y-1) &= 1 - \prod_{i=1}^{n} q_i^{y+1} - 1 + \prod_{i=1}^{n} q_i^{y} \\
&= \prod_{i=1}^{n} q_i^{y} - \prod_{i=1}^{n} q_i^{y+1} \\
&= \prod_{i=1}^{n} q_i^{y} \left(1 - \prod_{i=1}^{n} q_i\right) = Q^y(1-Q) = Q^y P
\end{aligned}$$

9. Show that $E|X| \geqslant |E(X)|$.

$$|E(X)| = \left|\sum_{i=1}^{\infty} x_i \, pi\right| \leqslant \sum_{i=1}^{n} \mid x_i \mid p_i = E \mid X \mid$$

10. If $X \backsim \text{NB}\left(1995, \frac{1}{2}\right)$, find $P(X \leqslant 1994)$.

$$= I_{1/2}(1995, 1995) = -\frac{1}{2}$$

11. Six identical dice thrown independently, S = number of dice showing even face $V(S)$ = ?

$\text{Bin}\left(6 \cdot \frac{1}{2}\right)$. $\left[P(\text{ even face appear }) = P(\text{ odd face appear }) = \frac{1}{2}\right]$.
$V(S) = npq = 6 \times \frac{1}{2} \times \frac{1}{2} = \frac{3}{2}$.

12. Three independent throw of a dice (fair). It X = number of faces showing 6. Find $E(3-X)^2$.

$$X \sim \text{Bin}\left(3, \frac{1}{6}\right), \quad Y = (3 - X)$$
$$Y \sim \text{Bin}\left(3, \frac{5}{6}\right).$$
$$E\left(Y^2\right) = \frac{2}{3}.$$

13. A box contain 2^n tickets among which $\binom{n}{i}$ tickets are no 1. m tickets are drawn from the box. What is the expected sum of these drawn tickets?

X_i = The no on the ith selected ticket.

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x^1 + \ldots + \binom{n}{n}x^n$$

$S = X_1 + X_2 + \ldots + X_m = \sum_{i=1}^{n} E\left(X_i\right)$.

$$n(1+x)^{n-1} = \binom{n}{1} + 2^n\binom{n}{2} + \cdots + nx^{n-1}\binom{n}{n}$$

$$E\left(X_i\right) = 0\frac{n_{C_0}}{2^n} + 1 \cdot \frac{n_{C_1}}{2^n} + \cdots + n\frac{n_{C_n}}{2^n} = \frac{1}{2^n}\sum_{i=1}^{n} i \quad n_{C_i} = \frac{n2^{n-1}}{2^n} = \frac{n}{2}$$

14. A box contain a white b black ball. Let c balls are drawn. Find expected value of the no of white balls. (with replacement draw).

(i) $X \sim \text{Bin}\left(c, \frac{a}{a+b}\right)$.

$$E(X) = \frac{ca}{a+b}$$

(ii) Let $X_i = \begin{cases} 1 & \text{white ball drawn at ith draw.} \\ 0 & \text{black ball drawn at ith draw.} \end{cases}$

$$E(S) = E\left(\sum_{i=1}^{c} X_i\right) = \sum_{i=1}^{c} E\left(X_i\right) = \sum_{i=1}^{c}\left(1\frac{a}{a+b} + 0 \cdot \frac{b}{a+b}\right)$$
$$= \frac{ca}{a+b}$$

15. Prove:

$E(\min(X,Y)) \leqslant \text{Min}(E(X), E(Y))$
$E(\max(X,Y)) \geqslant \text{Max}(E(X), E(Y))$
$\text{Min}(X,Y) = \frac{1}{2}[(X+Y) - |X-Y|]$.
$\text{Max}(X,Y) = \frac{1}{2}[(X+Y) + |X-Y|]$.
$|E(X-Y)| \leqslant E|X-Y|$
$-E|X-Y| \leqslant -|E(X) - E(Y)|$
$\frac{1}{2}\left[-E|X-Y| + E(X) + E(Y)\right] \leqslant \frac{1}{2}[E(X) + E(Y) - E|X| - E(Y)|]$
$\Rightarrow E\min(X,y) = \frac{1}{2} = \frac{1}{2}\left[E(X) + E(Y) - E(X-Y)\right] \leq \frac{1}{2}[E(X) + E(Y) - |E(X) - E(Y)|]$
$\Rightarrow \min[E(X), E(Y)]$